

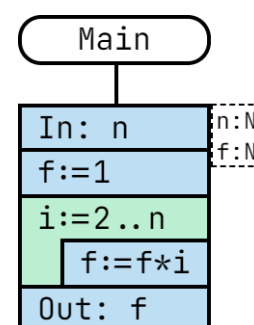
Task 24-25 solutions

Create a specification (<https://progalap.elte.hu/specifikacio/>) and algorithm (<https://progalap.elte.hu/stuki/>) for the tasks below!

24. Let's write a program that calculates the factorial of n !
For example: $5 \rightarrow 120$ (5 factorial = $5! = 1*2*3*4*5 = 120$)

Specification for task1

In: $n \in \mathbb{N}$
Out: $f \in \mathbb{N}$
Pre: $n \geq 1$
Post: $f = n!$
 Pt2: $f = 1 * 2 * \dots * n$
 Pt3: $\forall i \in [1..n]: (f = f * i)$
 Pt4: $f = 1$ and $\forall i \in [2..n]: (f = f * i)$
 Pt5: $\prod_{1..n} i$

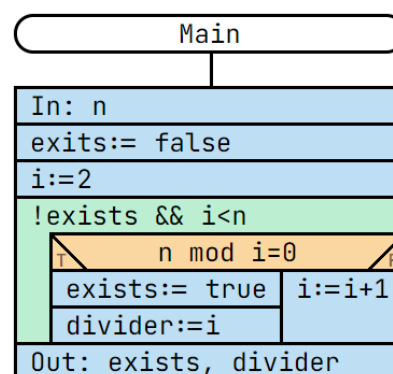


Algorithm for task1

25. Provide a proper divisor of a positive integer! (The proper divisors of a number are its divisors excluding 1 and the number itself.)
For example: $42 \rightarrow 6$ (but 2, 3, 7, 14, 21 are also valid), while $13 \rightarrow$ none.

Specification for task2

In: $n \in \mathbb{N}$
Out: $\text{exists} \in \mathbb{L}, \text{divider} \in \mathbb{N}$
Pre: $n \geq 1$
Post: $\text{exists} = \exists k \in [2..n-1]: ((n \bmod k) = 0)$
 and $\text{exists} \rightarrow ((n \bmod \text{divider}) = 0)$ and $2 \leq \text{divider} \leq n-1$



Algorithm for task2